

Week 12 Solutions

A promise problem: Consider a function $f : \{0, 1\} \rightarrow \{0, 1\}$ and a task whereby an algorithm must output if f is constant or balanced. (A function is ‘constant’ if it returns the same value $f(x)$ for all x . It is ‘balanced’ if it returns 0 for half the possible values of x , and returns 1 for the rest.)

(a) Given an oracle $U_f(|x\rangle|y\rangle) = |x\rangle|y \oplus f(x)\rangle$ and the query $|+\rangle$, show that oracle U_f returns,

$$U_f|+\rangle = \begin{cases} \pm|+\rangle, & \text{if } f \text{ is constant,} \\ \pm|-\rangle, & \text{else if } f \text{ is balanced.} \end{cases} \quad (1)$$

(a) *Answer:*

Substitute the identity for $|+\rangle$ and expand using linearity,

$$U_f|+\rangle = \frac{U_f}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle), \quad (2)$$

$$= \frac{1}{2}(U_f|00\rangle - U_f|01\rangle + U_f|10\rangle - U_f|11\rangle). \quad (3)$$

By definition of the oracle U_f , this gives

$$U_f|00\rangle = |0\rangle|0 \oplus f(0)\rangle = |0\rangle|f(0)\rangle \quad (4)$$

$$U_f|10\rangle = |1\rangle|0 \oplus f(1)\rangle = |1\rangle|f(1)\rangle \quad (5)$$

$$U_f|01\rangle = |0\rangle|1 \oplus f(0)\rangle = |0\rangle|\neg f(0)\rangle \quad (6)$$

$$U_f|11\rangle = |1\rangle|1 \oplus f(1)\rangle = |1\rangle|\neg f(1)\rangle \quad (7)$$

First consider only the constant case $f(x) = 0, \quad \forall x$. Substituting this into the oracle expressions gives,

$$U_f|00\rangle = |0\rangle|0\rangle = |00\rangle \quad (8)$$

$$U_f|10\rangle = |1\rangle|0\rangle = |10\rangle \quad (9)$$

$$U_f|01\rangle = |0\rangle|\neg 0\rangle = |01\rangle \quad (10)$$

$$U_f|11\rangle = |1\rangle|\neg 0\rangle = |11\rangle \quad (11)$$

$$\implies U_f|+\rangle = |+\rangle, \quad \text{if } f(x) = 0, \quad \forall x. \quad (12)$$

Now consider the constant function $f(x) = 1, \quad \forall x$ and substitute $f(0) =$

$f(1) = 1$ into the oracle.

$$U_f|00\rangle = |0\rangle|1\rangle = |01\rangle \quad (13)$$

$$U_f|10\rangle = |1\rangle|1\rangle = |11\rangle \quad (14)$$

$$U_f|01\rangle = |0\rangle|\neg 1\rangle = |00\rangle \quad (15)$$

$$U_f|11\rangle = |1\rangle|\neg 1\rangle = |10\rangle \quad (16)$$

$$\implies U_f|+-\rangle = \frac{1}{2}(|01\rangle - |00\rangle + |11\rangle - |10\rangle), \quad (17)$$

$$= \frac{-1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle), \quad (18)$$

$$= -|+-\rangle, \quad \text{if } f(x) = 0, \quad \forall x. \quad (19)$$

In the second last line, we are re-arranging terms, and the last line follows from the definition of $|+-\rangle$. Notice that this constant function is a negation of the first constant function, and it gives us a -1 overall but the quantum state doesn't change under U_f .

Next consider the balanced function $f(0) = 0, f(1) = 1$. Substituting this into the oracle expressions gives,

$$U_f|00\rangle = |0\rangle|0\rangle = |00\rangle \quad (20)$$

$$U_f|10\rangle = |1\rangle|1\rangle = |11\rangle \quad (21)$$

$$U_f|01\rangle = |0\rangle|\neg 0\rangle = |01\rangle \quad (22)$$

$$U_f|11\rangle = |1\rangle|\neg 1\rangle = |10\rangle \quad (23)$$

$$\implies U_f|+-\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |11\rangle - |10\rangle) \quad (24)$$

$$= \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle) \quad (25)$$

$$= |- - \rangle, \quad \text{if } f(0) = 0, f(1) = 1. \quad (26)$$

Again, in the second last line, we are re-arranging terms, and the last line follows from the definition of $|- - \rangle$.

The last balanced function, $f(0) = 1, f(1) = 0$, is the negation of the first balanced function (above) and this will only give us a over sign difference. We can deduce that $U_f|+-\rangle = -|- - \rangle$, if $f(0) = 1, f(1) = 0$.

Putting it all together, we have shown,

$$U_f|+-\rangle = \begin{cases} \pm|+-\rangle, & \text{if } f \text{ is constant,} \\ \pm|- - \rangle, & \text{else if } f \text{ is balanced.} \end{cases} \quad (27)$$

(b) Using the oracle U_f in (a), an algorithm outputs $|D\rangle$,

$$|D\rangle := (H \otimes H) U_f |+-\rangle. \quad (28)$$

Show that

$$|D\rangle = \begin{cases} \pm|01\rangle, & \text{if } f \text{ is constant,} \\ \pm|11\rangle, & \text{else if } f \text{ is balanced.} \end{cases} \quad (29)$$

(b) *Answer:*

Substitute the oracle from (a) and use the identities provided,

$$|D\rangle = \begin{cases} (H \otimes H) U_f |+-\rangle = \pm (H \otimes H) |+-\rangle = \pm H|+\rangle \otimes H|-\rangle = \pm|01\rangle, \\ (H \otimes H) U_f |+-\rangle = \pm (H \otimes H) |--\rangle = \pm H|-\rangle \otimes H|-\rangle = \pm|11\rangle, \end{cases} \quad (30)$$

(c) How many queries does (b) need to solve this problem. Explain why.

(c) *Answer:* We can read from the question (b) itself that it will take only 1 query. The signs $\pm$ do not matter since the probability of measuring an outcome only depend on squared values $|\pm|^2 = 1$ (Born rule). If the function is constant, we measure 0 on the first qubit with 100% Born probability. If the function is balanced, we measure 1 on the first qubit with 100% Born probability.